

Sri Datta Ganesh Bandreddi

sbandred@andrew.cmu.edu • (412) 284-8386 • [linkedin.com/in/sri-datta-bandreddi](https://www.linkedin.com/in/sri-datta-bandreddi) • <https://cosmobeans.dev>

EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

PHENOM

Hyderabad, India

Product Development Engineer (Enterprise Graph Team)

Dec 2023 - Jul 2025

- Led a team of six engineers to re-architect a multi-tenant AI analytics platform, enhancing scalability and reliability under concurrent tenant load by refactoring shared services and reducing cross-tenant coupling
- Scaled production knowledge graph to 1M+ entities with continuous ingestion, by designing validated aggregation pipelines integrated with Neo4j graph schemas and Iceberg versioned storage
- Ensured high availability for production workloads serving 100+ enterprise clients, by managing distributed Kafka, PostgreSQL, and Redis clusters and building CI/CD pipelines using Jenkins and Helm charts on Kubernetes
- Accelerated AI-driven query response times by 40% across 100+ clients, by tuning PostgreSQL horizontal sharding strategies

BYJU'S

Hyderabad, India

Software Engineer (Core Platform Team)

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% while maintaining reliability for 1M+ users, by deploying scheduled AWS ECS auto-scaling policies and maintaining Infrastructure-as-Code (IaC) configurations based on user traffic patterns
- Eliminated dependency on external identity services by 50%, by leading development of an in-house authentication and authorization platform using Go and Java (Spring Boot)

PHENOM

Hyderabad, India

Intern (Machine Learning)

May 2021 - Jun 2022

- Boosted recommendation accuracy by building persona classification models using NLP techniques (spaCy, BERT embeddings) across 3 data sources, processing user behavioral signals at scale to serve personalized content for production recommendation pipelines
- Accelerated recommender system feature preprocessing by migrating CPU-based data pipelines to NVTabular on GPU, reducing preprocessing latency and validating scalability improvements across 2 production databases

PROJECT EXPERIENCE

Research Project, Efficient Attention Mechanisms (Carnegie Mellon University) ([link](#))

Aug 2025 - Present

- Validated $> 200\times$ speedups and sub-quadratic memory scaling for linear attention mechanisms, by implementing and benchmarking on Long Range Arena tasks at 1k-8k token sequence lengths
- Profiled multi-GPU training with NCCL across FP8/BF16 configurations using Nsight Systems on Slurm-managed clusters, identifying optimal precision-performance tradeoffs for attention variants
- Authored and submitted a research paper to ICML 2026, demonstrating runtime, memory, and accuracy tradeoffs across 5+ efficient attention architectures benchmarked on vision and diffusion tasks

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Achieved 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification outperforming ResNet-50, EfficientNet-B3, and ViT baselines, by training a Swin Transformer scaled to 210M parameters with fused attention kernels
- Optimized inference latency by $3\times$ over PyTorch eager execution, by converting trained models to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and deploying on Triton Inference Server with adaptive request batching

EMG Hand Gesture Recognition, Neural Networks (Carnegie Mellon University)

Sep 2025 - Oct 2025

- Designed neural network models for EMG-based hand gesture classification, achieving 85-92% accuracy on 16-channel, 1000 Hz signals, using PyTorch LSTM/CNN architectures with attention mechanisms
- Performed controlled ablation studies across CNN, LSTM, and attention-based architectures, validating results over subject-wise splits and multi-channel EMG configurations to identify generalizable inductive biases

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B parameter LLM with stable convergence and $\sim 40\%$ reduced memory footprint on $4\times$ H100 GPUs, by configuring tensor and pipeline parallelism with NCCL-backed communication, gradient checkpointing, and using LoRA/QLoRA adapters
- Achieved $2.1\times$ training throughput improvement over naive data parallelism, by tuning NCCL parameters, gradient bucketing, and optimizer state sharding across GPUs